

**Theme - 2 City Intelligence:** Build software, AI, maps, and data tools that turn Lahore's open and existing data into a live, predictive, and citizen-facing view of the city.

## Theme Overview

The cities that built real intelligence into their operations did it by connecting data that already existed, not by inventing new infrastructure. Rio de Janeiro's Centro de Operações pulls live feeds from dozens of agencies, hundreds of cameras, and a network of rain gauges into a single operational view, so that when rainfall crosses a dangerous threshold, warnings reach at-risk neighbourhoods within seconds. Singapore's WaterWiSe network can locate a leaking pipe anywhere on the island within a day. Neither system depended on new technology so much as on breaking down the silos between departments that already held the data.

Lahore has many of the same underlying pieces. Suthra Punjab GPS-tracks its collection fleet and geo-tags worker photographs. Punjab EPA operates air quality monitors. The Punjab Information Technology Board is building One Map Punjab to unify land and geographic records. What Lahore does not yet have is a system that connects any of this into one live view, predicts problems before they become visible, or lets citizens contribute verified data of their own. That gap, siloed data with no predictive or citizen-facing layer, is what this theme asks you to close.

There is an honest constraint to name up front. Most of those official systems are not yet open to outside developers. Suthra Punjab's fleet data and PITB's land records are internal, and EPA's monitoring is not published as a clean, downloadable feed. So this theme works on two tracks at once. You build your prototype on the open data that genuinely is available today, and you design it so that an official feed could plug in later without a rebuild. The government systems above are your inspiration and your picture of what good looks like. The open datasets listed under each problem statement are what you actually build on.

## What You'll Be Working With

We encourage you to work with open datasets for creating solutions under this theme. Some examples of open datasets we can share are live air quality from [OpenAQ](#) and [AQICN](#), weather and rainfall from [Open-Meteo](#), the city's roads and facilities from [OpenStreetMap](#), population from [WorldPop](#), census and demographic figures from the [Pakistan Bureau of Statistics](#) and [Open Data Pakistan](#), and satellite imagery from [Copernicus](#) and [Google Earth Engine](#), and also freely available datasets on Kaggle. Most free datasets cover all of Pakistan, so suggest trimming them to the Lahore or Punjab area you are working on. Full details and links are in the Open Data Sources section at the end.

## What We're Looking For

Under this theme, participants are expected to build practical, data-driven solutions that help governments, communities, and citizens better understand, predict, and respond to urban challenges.

Your solution should combine one or more open datasets with software, artificial intelligence, mapping, analytics, or visualization tools to demonstrate how data can improve public services and decision-making.

Your prototype does **not** require access to internal government systems. Instead, demonstrate your concept using publicly available datasets while ensuring your solution can scale to incorporate official government data in the future.

Successful solutions will typically:

- Combine multiple datasets to generate richer insights.
- Transform raw data into actionable information rather than simply displaying it.
- Present information through intuitive dashboards, interactive maps, AI assistants, or mobile/web applications.
- Help citizens, policymakers, or city officials make faster and better decisions.
- Be scalable so that additional government datasets can easily be integrated over time.

Teams are encouraged to think beyond traditional dashboards and explore innovative ways of making city intelligence more accessible and useful.

## Problem Statement 1. A Real-Time Nervous System for Lahore's Civic Data

**Solution type invited:** Software, a unified dashboard

**Background.** Global command centres like Rio's built a single operational view by connecting data that already existed across departments. Lahore's equivalent pieces, waste collection, air quality, and land records, exist as separate systems today, but none of them share a live, combined view.

**Current limitation.** Almost everything you might want to know about a neighbourhood exists somewhere, but it is scattered across different websites and file types. A citizen or planner has no single place to see the essentials together: how many people live there, where the nearest hospitals and schools are, how the roads connect, and what the air is like right now.

**Desired outcome.** A prototype dashboard that pulls two or more live data sources into a single map, showing what a small-scale civic command view could look like for one district or union council.

**Open data you can use.** Live air quality from [OpenAQ](#) and [AQICN](#). Weather and rainfall from [Open-Meteo](#), which needs no sign-up. Roads, hospitals, schools, and facilities from [OpenStreetMap](#), with ready-made Pakistan downloads on the [Humanitarian Data Exchange](#). Population from [WorldPop](#), and demographic and district figures from the 2023 census on [Open Data Pakistan](#) and the [Pakistan Bureau of Statistics](#). Design the dashboard so an official feed, such as EPA monitoring or a fleet-tracking layer, could slot in later, but demonstrate it on the open layers above.

**Ideas to get started.** A know your neighbourhood page with a simple search box and a printable summary. A map that colours each area by how far it is from the nearest hospital or school. A layered map where a user switches each kind of information on and off.

## Open Data Sources You Can Use

Every source below is free and public. None requires special government permission. Most national datasets cover all of Pakistan, so trim them to the Lahore or Punjab area you are working on.

### Data about Pakistan and Punjab

- [Open Data Pakistan](#) — More than a thousand datasets on health, the economy, education, and the environment, including census figures, Punjab Development Statistics, and a Lahore property listing dataset with map coordinates.
- [Pakistan Bureau of Statistics](#) — Official national data, including the 2023 census with district-level tables in PDF and Excel.
- [Bureau of Statistics, Punjab](#) — Provincial statistics on population, health, education, and the economy.

### Maps of Lahore

- [OpenStreetMap](#) — A free, community-built map of the city: roads, buildings, hospitals, schools, shops, and more.
  - [Humanitarian Data Exchange](#) — Ready-made downloads of Pakistan's OpenStreetMap data as map files you can open in common tools.
  - [HOT Export Tool](#) — Draw a box around Lahore and download exactly the map features you want.
  - [Geofabrik Pakistan](#) — The full Pakistan OpenStreetMap file, updated daily, for teams comfortable with larger datasets.
- **Air, weather, and environment**
    - [OpenAQ](#) — Free, developer-friendly access to air quality readings from public sensors.
    - [AQICN \(Lahore\)](#) — Live air quality index for Lahore stations, with a documented way to pull the data into your app.

- [Open-Meteo](#) — Free temperature, wind, and rainfall data, past and forecast, with no sign-up required.
- [Kaggle](#) — Free, accessible datasets published by users across the global.

## Satellite data

- [Copernicus Browser](#) — The European Space Agency's free gateway to satellite images of Lahore, useful for greenery, heat, air, and city growth.
- [Google Earth Engine](#) — Free for research and non-commercial use, it lets you analyse years of satellite data in your browser without downloading huge files.
- [WorldPop](#) — Detailed population maps that estimate how many people live in each small area, useful for showing who is affected by anything you map.